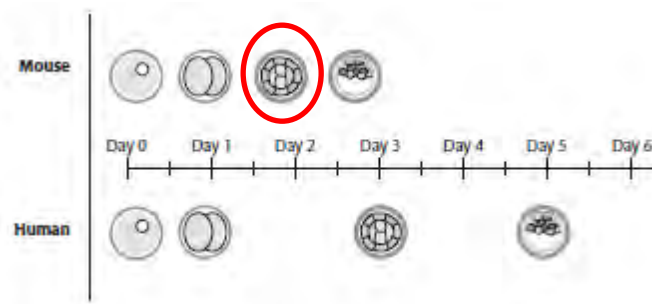


Question Number	Answer	Additional guidance	Mark
4(a)	A calculation showing the following steps: • correct difference (1) • correct percentage difference to 2 significant figures (1)	Mark the answer on answer line first <u>Example of calculation:</u> $170 - 105 = 65$ $(65 \div 105) \times 100 = 62\%$ Correct answer (62) with no working gains full marks Accept 38% for both marks	(2)

Question Number	Answer	Additional guidance	Mark
4(b)(i)	The only correct answer is B $2.85 \times 10^{-3} \text{ (mm}^3\text{)}$ <i>A is not correct because that is the volume in cm³</i> <i>C is not correct because that is not the volume in mm³</i> <i>D is not correct because that is the volume in μm³</i>		(1)

Question Number	Answer	Additional guidance	Mark
4(b)(ii)	 The diagram illustrates the timeline of early embryonic development for both mouse and human. A horizontal timeline from Day 0 to Day 6 is shown. Above the timeline, 'Mouse' development is depicted: a single-cell zygote at Day 0, a two-cell embryo at Day 1, a red-circled morula at Day 2, and a blastocyst at Day 3. Below the timeline, 'Human' development is depicted: a single-cell zygote at Day 0, a two-cell embryo at Day 1, a morula at Day 3, and a blastocyst at Day 5.	accept circle without morula label accept morula label without circle	(1)

Question Number	Answer	Additional guidance	Mark
4(b)(iii)	A description that includes the following points: • fusion of cortical granules with cell (surface) membrane and release of enzymes (1) • hardening of zona pellucida (to prevent polyspermy) (1) • fusion of two haploid nuclei (to form diploid nucleus) (1)	Accept joining of two haploid nuclei	(3)

Question Number	Answer	Additional guidance	Mark
4(c)	An answer that includes the following points: • it is a short distance between (mitochondria and lipid (1) • quicker diffusion of {lipid / fatty acid / glycerol} (into mitochondria) (1) • lipids used in {respiration / ATP production} (1)	accept they are close together Accept (lipid droplets) converted to glucose ignore energy storage	(3)